BRAC UNIVERSITY

CSE423 - Computer Graphics - Spring 2026

Assignment-2

Total Marks: 40

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1. Naimur Rahman Sifat, former world champion of the legendary "Sharadin Ghum" competition, has finally retired—ofcourse, after one last nap! Now, he's turning his focus to something slightly more demanding: Projection.

   Under the guidance of the renowned teacher,Professor Ashfaq Khan, Sifat is learning the art of mechanical drawing—starting with how to sketch the top view of a building. Not stopping there, he's also diving into projection matrices to figure out how to map a point onto the YZ-plane.

   a. Which projection mechanism was taught by Professor Khan? Justify your answer. **[1+1]**
   b. Derive a 4x4 projection matrix to map a point P(x,y,z) on the YZ-plane. **[5]**

2. Right now, Sifat is creating a technical drawing of a building where the front face of the object is shown in true shape and size, while the depth is represented along lines receding at 45° to the horizontal plane, and depth dimensions are kept at full scale.

   Identify which type of projection is being used here and justify your answer with two characteristics of this projection type. What would change in the drawing if the depth axis were scaled to half of its actual length? **[3+2]**

3. Suppose, Sifat has decided to perform an Orthogonal projection of a point P (17,−8,42) on the projection plane X = 26. Find out the new coordinate of point P after projection. **[2+1]**

4. a. Derive 4x4 general-purpose perspective projection matrix using appropriate figure, showing P, P', COP, PP, and **q. [4]**
   b. Using the general-purpose perspective projection matrix from 4(a), derive the projection matrix when COP is at origin. **[2]**
   c. Using the general-purpose perspective projection matrix from 4(a), derive the projection matrix when PP is at origin. **[2]**

5. Imagine, the FIFA World Cup 2026 trophy is brought to BRAC University as part of the event "Football e Jibon." The most enthusiastic football fan, Sadab Shawgat, captures a picture of the trophy.
   a. The coordinate of Sadab's eye is at (0,0,0), the projection plane is parallel to the XY-plane and it is 10 units far on the z-axis from his eye, a random point on the

trophy is P(25,10,12). After taking the picture, find out the coordinates of the projected point of the point P. **[4]**

b. Sadab was not happy with the picture he clicked and thus decided to change his position such that the origin is now at the projection plane. The distance between his eyes and the projection plane is 250 units and the projection plane is on the XY-plane. What will be the coordinates of the projected point of the point P in this setup? **[4]**

6. While studying late at night for the CSE 423 final, you realize it is gently raining outside through your glass window. The window is standing at z = -250. You look through the window down at a flower shop across the street, where a flower, for its unique colour, catches your attention. If a line segment is drawn starting from one side of the shop, (100, 100, -350), and ending at the other side (500, 80, -350), the flower's position, P, falls on this line segment's 80% of the way. From where you are standing, the x and y coordinates of your eye are (60, 200). The z distance between you and the glass window is 1/5th of the z distance between the window and the flower. Find the flower's projected coordinate, P', on the window. **[9]**